# Loco Stablecoin Infrastructure PRD


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## Executive Summary


Loco is transitioning from a beans-based virtual currency system to a USDT stablecoin infrastructure to enable seamless cross-border payments for streamers and viewers. This PRD outlines the complete implementation of crypto wallet functionality, treasury management, and user experience redesign.


## 1. Problem Statement

### Current State Issues

- **Beans complexity**: Creates confusion for internal teams and streamers
- **Currency arbitrage**: Fixed bean-to-local currency rates create arbitrage opportunities
- **Payment friction**: Traditional payment methods face significant friction in target markets
- **Scalability limitations**: Different bean conversions are needed for each country
- **Revenue leakage**: Streamers receive only 6-10% of viewer payments due to diamond-to-bean conversion

### Target Markets

- Turkey
- Argentina
- Brazil
- Indonesia
- Thailand
- Egypt
- Mexico
- Malaysia
- Vietnam
- South Korea

# 2. Solution Overview

## Core Components

1. **Phase 1**
    a. Crypto Wallet Infrastructure (Fern Crypto integration)
    b. Streamer wallet Interface Redesign
    c. Treasury Management System
    d. KYC
    e. Payment Flow
    f. Beans Migration Strategy
    g. Optimisation
2. **Phase 2**
    a. Automated CMP earnings (Retool)
    b. Commissions deduction system (Retool)
    c. Notification System
    d. Reporting structure for reconciliation

## Key Benefits

- Reduced cross-border transaction costs
- No revenue loss while converting beans to local currency for the streamers
- No requirement to partner with multiple PSPs across the markets leading to higher integration time and complicated regulations
- Faster settlements
- Consistent user experience across markets
- Elimination of currency arbitrage
- Direct USDT earnings for streamers

## 2.3 Key use-cases Loco wants to solve:

1. **Transaction handling**
    a. Loco's Treasury: for settling transactions
    b. Automated payouts: Streamlined revenue distribution to streamers
    c. Off-ramp capabilities: Conversion to local currencies with competitive rates

2. **User Onboarding**
    a. Custodial wallet creation: Seamless, programmatic wallet generation for new users

3. **API-based fund transfers to external wallets**:
    a. We envision a setup similar to Net Banking where users can log in on Loco's UI and transfer funds to an external wallet, be it their crypto wallet or fiat bank account

b. Technical Requirements
    i. API-based fund transfer
    ii. Webhook support for transaction notifications

# 3. Technical Architecture

## 3.1 Wallet Infrastructure (Vendor Name - Fern - documentation link)

**Wallet Creation Flow**

1. Once the streamer hits their first payment, we start showing the wallet page with the relevant transaction history and a CTA to “Complete KYC”
2. Clicking on the aforementioned CTA triggers a flow to add link either email or phone number based on the missing information about the streamer - this is a mandatory step
    a. In case streamer has multiple accounts with the other mentioned detail, we merge the two streamer accounts with higher tier account being retained
3. Once we receive the missing detail, we create the streamer's customer on Fern and we receive the KYC/B link.
4. Once the streamer hits their first payment (either diamond stickers or monthly contract earnings) & streamer reaches Master Tier, we create a custodial wallet for the streamer
5. We then ask users to add either their bank or crypto details for settlements
6. Please note: *We shouldn’t create a wallet for every streamer because that would result in multiple inactive wallets. At the point when the customer (i.e., streamer) is created in Fern's system, and this KYB/KYC link is generated. The merchant can only actually start to initiate payouts to fiat once the KYB/KYC is passed and their bank account status is active (usually takes anywhere between 10 mins to 24 hours for approval)*

**Wallets**: Manage treasury & streamer wallets

- Treasury Wallet (On-Ramp funds in USDT/USDC from Loco's Bank A/C)
- Streamer Wallet:
    - Programmatically create a wallet for every eligible Streamer (has earnings on the platform)
    - Allow streamers to access wallet balance from Loco's Streamer Dashboard (Wallet not accessible off-platform)

**Payment Processing:** Move funds between Wallets

- Programmatically move funds between Loco's Treasury Wallet and Streamer Wallets based on daily/weekly/monthly earnings via (Tipping, Subscription & Creator Monetization Program)
- Transaction details & history for each wallet available via API
- Webhook support for transaction notifications

**Off Ramp (Fiat) / Off-Platform Transfer (StableCoin)**:

- Off-Platform Transfer: Streamers can initiate a fund transfer from their Loco wallet to another wallet off-platform (Off Ramp in Stable Coin to their personal wallet)
- Off Ramp (Fiat):
    - Off Ramp to allow Streamers to withdraw funds from their Loco wallet to their local bank accounts. [Streamer Initiated / Automated / Programmatic]
    - Connect & verify bank account to Loco wallet
- Transaction / Settlement details into their bank accounts or Wallet available via API.

**KYC** (we have to go with option 1 because within the builder model, Fern doesn't allow to customize KYC form natively)

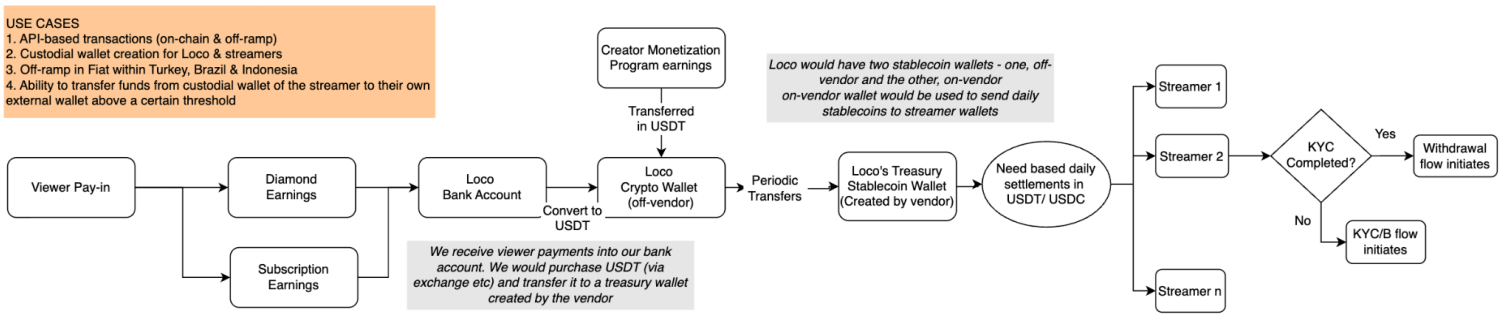
## 3.2 Treasury Management

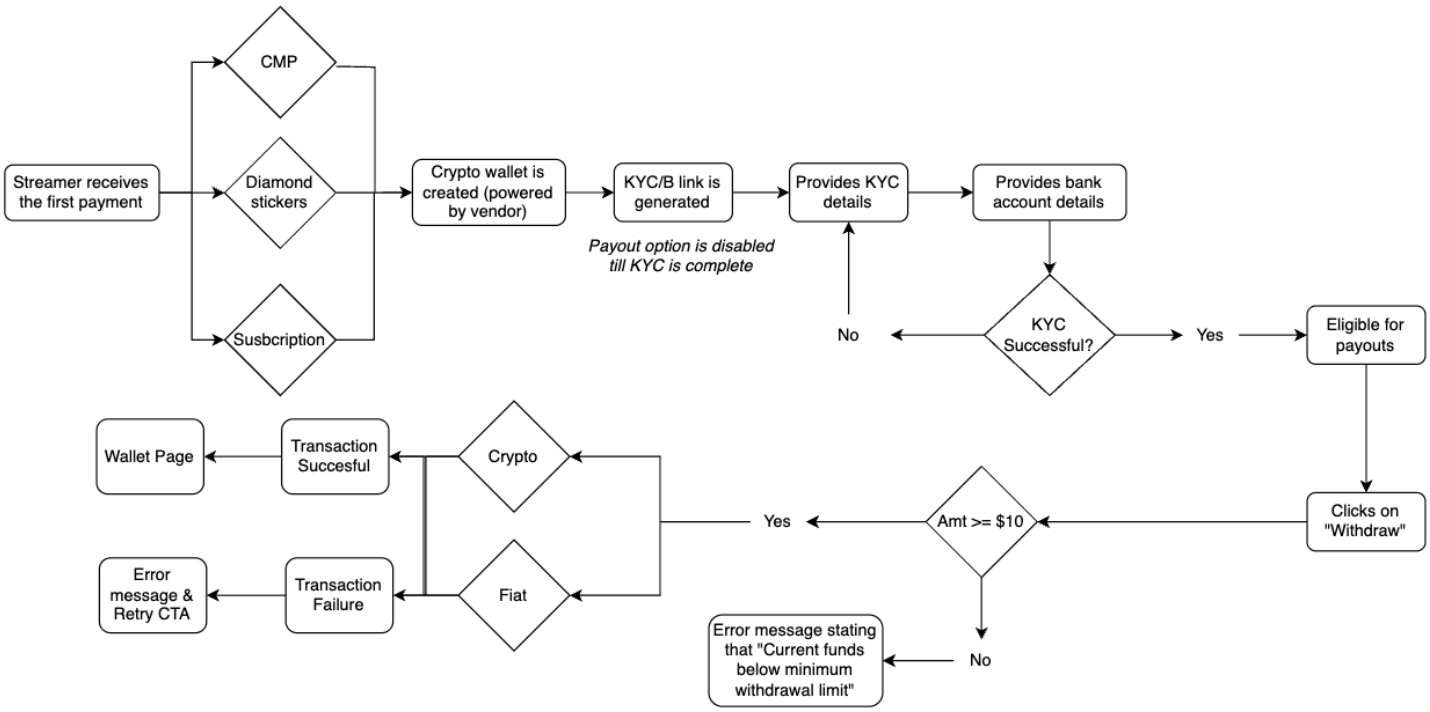
### 3.2.1 Money Flow Architecture

**Current Flow Problems**:

- Viewers pay $0.18-$0.29 per diamond
- Streamers receive only 1 INR equivalent (~$0.012)
- 90%+ revenue leakage

**New Flow Design**:






### 3.2.2 Settlement Strategy

- Viewer payments through Stripe settle daily or weekly or monthly (as per the choice made on the dashboard) & Platform billing settle in 30 days to Loco's bank account
- Finance team tops up Loco's nodal wallet with weekly frequency
- Loco's Nodal wallet is used to complete settlement for streamers for earnings they had
    - For diamonds, as the viewer donates it to the streamer, we need to transfer equivalent amount to the streamer from Loco's fern wallet; this can happen at a delay of 5 mins or so
    - For subscriptions, as the viewer buy subscriptions for a streamer, equivalent amount to the streamer from Loco's fern wallet; this can happen at a delay of 5 mins or so
    - For CMP, automatically gets transferred to the streamer wallet from Loco's fern wallet at the pre-set day of the CMP earnings
- Before transferring the amount to the streamer's wallet, we deduct any expenses that we might want to (need to build retool app to keep configuring this - with access to the pod leads and product manager)

### 3.2.3 Treasury Components

**Loco Treasury Wallets**:

- **Operations Wallet**: Handles gas fees and operational costs
- **Reserve Wallet**: Maintains liquidity buffer

- *Finance team figuring out a mechanism to buy stablecoins that can be transferred to our Fern x Loco nodal wallet*

#### 3.2.4 Fallback Mechanism

- In case Fern's services are down, we notify users that the system is down and come back later to retry

#### 3.2.5 Geo & daily transfer restriction for stablecoin payments

- We should have the ability to restrict out of platform stable coin transfers by country (where the streamer is based) - to be done via retool
- Maximum Daily Withdrawal Limit at the user level of $3000 + monthly of $50,000 (this should be configurable via Retool); The timezone would be UTC for daily & monthly limits; for all the other transactions, we would follow complete withdrawals, however, if the user has more than $3000, we allow partial withdrawal i.e. $3000 instead of the actual amount the streamer has.

### 3.3 Diamond Value Standardization for streamer payouts

**Current Issues**

- Diamond prices vary from $0.18 to $0.29 across packs
- Streamers receive fixed 1 INR regardless of purchase price
- Impossible to track diamond source

**Solution**

- **Fixed Diamond Value**: 1 Diamond = $0.14 USD (configurable via retool)
- **Calculation**: Lowest diamond price (after deducting platform commissions; across all packs on both the platforms) is $0.14
- **Global Standard**: Consistent across all markets

Review ➕ Diamonds pricing for viewer & streamer for more details

## 4. Feature Specifications

### 4.1 Streamer Wallet Interface

#### 4.1.1 Dashboard Redesign

**Current Display**: Beans balance

**New Display**:

- **Primary Balance**: Current USD balance (prominent)
- **Diamonds**: Show as USD equivalent (1 Diamond = $0.14)
- **Monthly Earnings**: Display in USD
- **Subscriptions**: Display in USD with dashboard link
- **Gold for streamer:** Gold amount that is in their wallet

#### 4.1.2 Transaction Ledger

**New Ledger Screen Requirements**:

- **Inflow Transactions**:
  - Diamonds credit (quantity & USD amount)
  - Subscription/gifting earnings (USD)
  - Monthly contract earnings (USD)
  - Gold
- **Outflow Transactions**:
  - Diamond redemptions (USD)
  - USD withdrawals
  - Platform deductions (fraud, commissions)
  - Gold

**Ledger Fields**:

- Transaction ID
- Description
- Type (Credit/Debit)
- Amount (USDT)
- Source/Destination
- Timestamp
- Status

### 4.1.3 KYC

- Once the wallet is created, Fern will send a URL for KYC.
- In this partnership with Fern, the KYC would happen via the third party vendor
- The design team can use this sample video link for creating the KYC flow.
- Existing Figma design for reference: link
- Input fields required:
  - Name
  - Date of birth
  - Address
  - Nationality
  - Proof of identity document
  - Proof of address document
  - Employment status
  - Source of funds

        - Intended volume use for the account

### 4.1.4 Bank details

- Post completion of KYC, a bank form will be generated to add the bank details.
- Use the current input fields to add bank details
- We can only have one bank account at a time - so if the the user has to add another bank account, he would need to delete the previous one

### 4.1.5 Transferring the USDT

- Once the KYC is completed, streamer becomes eligible for payouts
- Streamer can do the transfer in either their stablecoin (external) wallet or do a Fiat transfer to their bank account
- Post the selection, a request will be generated
    - On successful/failed/pending transfer Fern webhook will return the status
    - The payout fees will be deducted during the transaction itself - therefore, it's important to make users aware of the fees attached
- On wallet transfer:
    - Streamer can choose which network/chain they want to use
    - These networks can be displayed or hidden based on transaction fees, and can also be sorted in ascending order, with the lowest fees shown at the top
        - We will choose to avoid Ethereum because of the high costs (>~$2)

#### 4.1.6 External crypto wallets of the users

- Maximum external wallet crypto addresses that we will store at our end - 5 (configurable by the retool dashboard)
- Therefore, we will need to provide delete & add options to the user so that they can manage their list accordingly

### 4.1.7 Tutorial Video / FTUE

- A small tutorial video will be required in the wallet section to educate streamers
    - This video will be helpful for first-time crypto users.
        - Since crypto will be new for a lot of users
        - This will contain how to create a wallet, how to transfer the amount to either a wallet or a bank account.

### 4.1.8 Wallet Creation Flow

1. Customer Creation:
    a. A customer account will be created once a streamer has completed their first stream.
    b. As part of this process, providing an email address will be mandatory. This is a change from the current design, where email linkage is not required.

    c. Previously, the logic enforced that if a user signed up via phone number, then email verification was compulsory (and vice versa). That requirement was removed but will now be reinstated at the signup stage.
    d. For existing streamers, the migration or update path will be defined and communicated once the approach is finalized.

2. Wallet Creation:
    a. A wallet will be created for a streamer only after they receive their first transaction and have opted into CMP, i.e once they achieve Master tier along with their 1st payment (diamonds / subscriptions / CMP, etc).

3. Handling of Earnings:
    a. All funds from transactions will initially be stored inside the platform wallet until they are in Master tier(If they are in Organic i.e. not enrolled in CMP or Rookie tier it will be stored in the backend but it will continue to be shown in the ledger and wallet UI).
    b. These earnings will not be transferred to the streamer's external account until they achieve the Master tier. Upon reaching Master tier, the accumulated balance will be transferred to their wallet promptly.

4. UI Ledger:
    a. Streamer will be able to see how much and from where they have received the fund but that will be an unsettled amount, they have to complete all the conditions mentioned(Complete KYC, reach Master tier and have received any fund) earlier to get it deposited into their crypto wallet.
    b. For streamer payouts i.e. when they are taking money out, we need to show transaction status, time stamp, account number they are withdrawing to and transaction hash / settlement UTR. (Fern doesn't provide settlement UTRs / Hash IDs at the moment but it's in the product pipeline)

5. Withdrawal Conditions:
    a. To withdraw funds, the following conditions must be satisfied:
        i. The streamer's wallet must already be created.
        ii. KYC verification must be successfully completed.
        iii. The streamer must have achieved the Master tier.
        iv. The available balance must meet the minimum withdrawal threshold.

## 4.2 Automated Earnings System

### 4.2.1 Creator Monetization Dashboard

**Current Process**: All the payments are calculated in Beans

**New Process**: Beans calculation to be changed to USDT

**Requirements**:

- API update
- Contract value input changes in retool
- Automated USD calculation update
- Month-end earnings display in USDT (Currently in Beans)

#### 4.2.2 Commission System

**Phase 1**: 100% earnings pass-through

**Phase 2**: Configurable commission deduction

**Configuration Parameters**:

- Streamer tier (from creator monetization dashboard)
- Country/geography
- Commission percentage
- Deduction transparency

**Technical Implementation**:

- Retool dashboard for commission configuration
- Real-time commission calculation
- Clear communication to streamers about deductions

### 4.3 Withdrawal System

#### 4.3.1 Dual Withdrawal Options

**Fiat Off-ramp**:

- Integration with Fern Crypto off-ramp
- Local currency support per market
- KYC verification required (Fern to take care of this via a link)
- Bank account linking

**Crypto Withdrawals**:

- Wallet-to-wallet transfers
- External wallet address validation
- Gas fee calculation and deduction
- Transaction confirmation system

#### 4.3.2 Withdrawal Configuration

**Minimum Thresholds**: $10 (Configurable via retool dashboard)

**Maximum withdrawal requests** - One per streamer per day (Configurable via retool dashboard)

## 4.4 Notification System

### 4.4.1 In-App Notifications

- Transaction confirmations
- Earnings updates
- Balance changes
- Withdrawal status updates

### 4.4.2 Push Notifications

- Successful withdrawals
- Failed transactions with reason codes
- Earnings milestones

### 4.4.3 Email Notifications

- Weekly earnings summary
- Monthly transaction statements
- Withdrawal confirmations

### 4.4.4 Slack alerts

- Daily notification at 00:00 UTC about
    - Balance in Fern's loco wallet
    - Balance in all the streamer wallets that haven't been redeemed
    - Total withdrawals made in the last day from the streamer wallets

# 5. Payment Integration

## 5.1 Viewer Payment Methods

**Current Methods (Maintained)**

- **Global (non-Brazil)**: Stripe (credit/debit cards)
- **Brazil**: PagBrasil (PIX, debit cards)
- **Mobile Apps**: Google Play/App Store billing

**New Methods (Phase 2)**

- **Crypto Payments**: Wallet-to-wallet transfers
- **On-ramp Solutions**: Fiat-to-crypto conversion

- **QR Code Payments**: Direct USDT transfers

**Currency Display**

- Base pricing in USD
- Local currency conversion based on browser location
- Dynamic exchange rates (captured at the end of the day)

# 6. Migration Strategy

## 6.1 Beans Phase-Out

**Migration Timeline**

- **Phase 1**: Parallel system operation
- **Phase 2**: Existing bean balance conversion and storing the data on our backend
- **Phase 3**: Complete beans removal

**Bean Conversion Process**

1. Calculate total bean balances per streamer by country
2. Apply historical bean-to-local currency rate across countries
3. Maintain transaction history and data for reconciliation & in case streamer reaches out again

# 7. Reports for reconciliation & data tracking *(to be picked up in Phase 2)*

We need to create 4 reports (refer to the sheet here to understand the attributes and format):

1. Summary Report
2. Loco wallet statement report
3. Wallet balance report
4. Master ledger report

# 8. Implementation Timeline

## Phase 1: Foundation (Weeks 1-4)

- Fern Crypto integration
- Beans decommission(Migration+CMP)
- Wallet creation system
- Treasury management

- Streamer dashboard redesign
- Transaction ledger
- Withdrawal system
- Internal testing

### Phase 2: Core Features (Weeks 5-6)

- Commission system
- Notification system
- Streamers' automated earnings via retool

# 9. Success Metrics

### Financial KPIs

- Transaction cost reduction
- Settlement time improvement

### User Experience KPIs

- Streamer wallet adoption rate
- Withdrawal success rate

### Technical KPIs

- Transaction processing time
- System uptime
- Error rates

# 10. Appendix

### 11.1 Open Questions for Finance team

- **Tax Compliance**: Tax reporting for crypto earnings?

# 11. FAQs (responses by the Fern team & general FAQs)

- Regarding Swift fees for B2B payments, on which transactions are these fees levied?
    - These will be applied whenever SWIFT is chosen as the destination payment method (indicating we're sending a SWIFT transfer on your behalf)
- Which stablecoins and networks do you support? Also, based on gas fees and reliability, could you suggest which chains/networks would be most suitable for our needs?

    - We support a broad set of stables, and can easily support more as desired. Today, our most popular stablecoins are USDC, USDT, and PYUSD. We support these networks: https://docs.fernhq.com/cryptocurrency-support Solana will be live next week. Based on gas fees and reliability, we'd recommend one of the L2s such as Base, Arbitrum, Polygon, or OP mainnet
- Could you please share the gas fees for the networks you support?
    - We don't currently charge gas for any L2s. If you're initiating transactions on Ethereum mainnet, we'll pass through gas costs at cost to you
- Do you offer automated payouts from wallets? If so, are there any thresholds that trigger these payouts, and can our streamers set these thresholds themselves?
    - This is a feature we're actively building! Should be live in a couple weeks. We support time-based payouts (e.g. every day, week, etc.) as well as threshold-based payouts (e.g. when a balance exceeds a set threshold)
- As referenced on your website, what exactly is the minimum transaction threshold, and does it apply to both on-chain and off-ramp transactions?
    - Minimum transaction size for crypto swaps is $0. Minimum transaction size for most fiat transfers is $1. For certain rails, we have more specifics:
        - BRL offramps: $10
        - SWIFT: $1000
        - NGN (on/offramp): $5
- What would be the wallet hosted network? and does it impact how & on which network users want to off-ramp their funds
    - Wallet Hosted Network: When you create a Fern wallet, we will create a safe multisig on all supported EVM chains. See (https://docs.fernhq.com/guides/create-fern-wallets) for more details. We will automatically bridge funds to the right destination chain your customer selects
- Will the balance change / wallet balances api show real time balance of the stablecoin value? Or it solely depends on our polling frequency?
    - Yes, the wallet reflects the balance on-chain (real-time)
- Can a user add multiple bank details and edit the existing bank details for fiat off-ramp
    - Yes, you can add multiple bank accounts. They will need to delete the bank account and create a new one
- Can we store crypto wallet addresses on our end?
    - Yes, you can store crypto wallet addresses on your end; It would use the create payment account endpoint with external crypto wallet type: https://docs.fernhq.com/api-reference/payment-accounts
- What will be the format of kyc and bank details - how would the URL be sent to us and how can we white label it?
    - The KYC link and bank link will be in the response on the API request (https://docs.fernhq.com/guides/create-and-verify-customers). However, the whitelabel is not available on the Builder Tier Model
- As I understand, the bank account addition step would happen outside of Loco. Can we whitelabel it and have it within Loco?

    - Bank account addition happens outside if you use the hosted form, but the payment account create endpoint with external bank account type can be used to easily add bank accounts via API
- When users want to edit their bank details or add a new bank account (by replacing the previous one), do we send them the same bank account form and it happens outside Loco?
    - Users cannot currently edit their bank account details, but they can delete them and create a new one. This can be done via the API
- Users would store their external crypto wallet addresses (for payouts) on Loco, right? or is there another form that they would have to fill to facilitate this?
    - The payment account create endpoint can be used to store the crypto wallet addresses inside Fern
- Will users be able to add multiple crypto wallet addresses, or are they restricted to just one similar to the bank account?
    - Users can have as many crypto wallet addresses as they want.
- During withdrawal, can users choose their preferred crypto network, or should Loco automatically recommend the best/lowest-cost network based on their wallet address?
    - Loco can recommend a network if they want, but users can choose any of the supported networks for withdrawal
- Is there an authentication step (e.g., OTP, 2FA) required before confirming a withdrawal? or the user can opt between a fiat / crypto payout and it automatically gets triggered based on the added details?
    - Authentication is done via api key, which Loco owns. Loco can implement more auth on user-based withdrawals
- What is this crypto wallet and how does it work?
    - Your new Loco wallet is a secure digital wallet where your earnings are stored in USDT (a stable cryptocurrency).
    - You can use the wallet to -
        - a. Withdraw your earnings from Loco to your bank account in local currency (flat withdrawal)
        - b. Transfer USDT to another crypto wallet of your choice.
- Can I use my wallet before completing KYC?
    - No. KYC is mandatory to make any withdrawals.
- When do I receive payments in my wallet?
    - Earnings are credited on a T+1 basis. This means that you will receive your earnings the next day.
    - For eg - Earnings from Monday will appear in your wallet on Tuesday and so on.
- How can I withdraw money from my wallet?
    - You have two options:
        1. Bank Transfer Convert your USDT to local currency and withdraw directly to your bank account.
        2. Crypto Transfer: Send USDT to any external crypto wallet. You can choose the network (like Ethereum or Tron) based on your preference.
- What if I entered the wrong wallet address for a crypto transfer?

    - Be very careful – crypto transfers are irreversible.
      If you enter the wrong address, the funds may be lost permanently. Always double-check the wallet address before confirming a transfer.
- What happens to my existing beans?
    - Your current Beans balance will be converted to USDT at the same value. This new balance will be visible in your wallet once the switch is complete.
- Do I need to pay any fees for withdrawals?
    - Yes. Transaction fees apply. Depending on the withdrawal method and network chosen, some fees will be charged.
- What happens if  I don't withdraw the USDT timely?
    - There is no time limit or expiry to the USDT. You can withdraw whenever you like, and your balance will keep adding up over time.
- Is there a minimum amount required to withdraw?
    - Yes - you'll need at least $10 worth of USDT in your wallet to initiate a withdrawal.
- Who do I contact if I face an issue with my wallet?
    - Please feel free to contact our support team for any issues.

# 12. Questionnaire

- When will the contract be signed?
    - Already got a legal update on this, will be closed soon. Prakhar is working on the contract signing part.
- When will the demo, tech and keys be shared?
    - Once the contract is signed, the keys will be shared.
    - Thursday will have a brief call with the engineering team to give an overview of the feature.
        - Post that they will read the documentation and will schedule a call with the Fern team for tech Q&A.
- How will the work be divided?
    - A single eng team will take complete ownership of it. It will be divided into 2 phases:
        - **Phase 1**
            - **Crypto Wallet Infrastructure** (Fern Crypto integration)
            - **User Interface Redesign**
            - **Treasury Management System**
            - **KYC**
            - **Payment Flow Optimisation**
        - **Phase 2**
            - **Beans Migration Strategy**
            - **Existing Beans conversion changes (Eg. Diamonds to beans, CMP)**
            - **Charges deduction system (Retool)**
            - **Notification System**

- How will Loco get the crypto?
    - Loco will buy the crypto from a vendor and will store it in their treasury.
    - From the reserve treasury, every week the amount will be added to the nodal wallet.
        - This will cost an extra gas/transaction fee on transfer.
        - Should use the reserve wallet only when 75% nodal wallet is empty, and top up only the nodal wallet for T+1 day transaction.
- When transferring the amount to Streamer, either for CMP or a diamond transaction, there will be some transactional charges involved. Who will bear those transaction charges?
    - Currently, Loco will bear those transaction charges.
    - Once we start taking the commission, we will add that % to the commission.
- How will the transaction charges be deducted?
    - Not sure: Ex: if we are transferring 100 USD worth USDT and say 1% charges are there, then 99 USD worth USDT will be transferred to the streamer's wallet
    - So we need to always calculate extra charges and send.
    - Or need to find a better solution for it.
- When transferring the amount to a bank or transferring it to another wallet streamer will have to bear the gas fee?
    - Yes streamer will bear the gas/transaction fee (Need to confirm with Firasat)
- How will the subscription revenue be transferred in the form of USDT?
    - Will discuss this with Amogh and Dinesh
- On which platform Loco's Nodal and Reserve wallet will be hosted?
    - Fern